



Péter Juhász

Quantitative Researcher

CONTACT

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EDUCATION

2022 – 2025 PhD in Mathematics
Aarhus University, Denmark

2019 – 2021 MBA in Finance
Budapest University of Technology and Economics, Hungary

2013 – 2016 MSc in Physics
Budapest University of Technology and Economics, Hungary

2013 – 2016 BSc in Physics
Budapest University of Technology and Economics, Hungary

SKILLS

- Stochastic Models
- Probability
- Data Analysis
- Python
- C++
- SQL
- GIT

LANGUAGES

- Hungarian
- English
- German

CERTIFICATIONS

- 2021 Software Architect ISAQB**
- 2021 Software Designer Bosch**

PROFILE

I am a PhD in Mathematics with experience in stochastic modeling, probabilistic analysis, and large-scale simulations with a background in Python/C++ model development. Previously, I was a machine learning researcher at Bosch, where I worked on predictive modeling, uncertainty estimation, data-driven decision-making, and scalable system design. I am interested in applying quantitative methods to forecasting and risk modeling in energy and financial markets.

PROFESSIONAL EXPERIENCE

PhD Researcher in Mathematics **2022 – 2025**
Aarhus University, Denmark

- Developed stochastic models and statistical tests for high-dimensional random systems.
- Designed and implemented high-performance Monte Carlo algorithms in C++ and Python for hypothesis testing and validation of models against datasets.
- Taught advanced probability, data analysis, and differential equations to mathematics and data-science students.

Machine Learning Researcher & Software Architect **2018 – 2022**
Bosch Group, Hungary

- Patented a Gaussian regression based trajectory prediction model, improving predictive accuracy by 3× and uncertainty calibration by 10× over baseline.
- Led a team of 11 developing, validating, and integrating environment prediction models for automated driving as a technical lead.
- Developed Python prototypes of prediction models to validate concepts before coordinating their integration.

Software Engineer **2016 – 2018**
Ericsson Telecommunications, Hungary

- Analyzed response times of distributed systems to optimize load-balancing strategies.
- Implemented performance-critical algorithms for Smart Services Routers.

AWARDS, GRANTS

2022 PhD Fellowship

Danish Data Science Academy
1 890 000 DKK

2021 Excellence Award

Hungarian National Bank
350 000 HUF

2025 Individual Travel Grant

European Mathematical Society
500 EUR

2024 Academic Mobility Grant

Aarhus University
20 000 DKK

SELECTED PUBLICATIONS, PATENTS

- C. Hirsch, B. Jahnel, and P. Juhász. **2025**
Functional limit theorems for edge counts in dynamic random connection hypergraphs
- P. Juhász. **2021**
Method and device for predicting the trajectory of a traffic participant, and sensor system
- P. Juhász. **2021**
Information propagation in stochastic networks
- P. Juhász, J. Stéger, D. Kondor, and G. Vattay. **2018**
A Bayesian approach to identify Bitcoin users